

Broadband Over Power Lines

Broadband over power lines (BPL) technology is a union of two applications in a single system. It uses the existing power grid hierarchy comprising a network of different transmission lines to carry high-speed digital data over a broad range of frequencies without causing any unreasonable significant interference. Thus, BPL holds great promise as a ubiquitous broadband solution that would offer a viable alternative to existing broadband technologies

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Despite the proliferation of broadband Internet technology in the last few years, many regions of the world, particularly rural and low-income areas, still do not have access to broadband services. Huge costs involved are the main obstacle to creating new infrastructure in such areas for existing broadband wired or wireless technologies like digital subscriber line (DSL), cable and satellite. But if broadband could be served through electric power lines, there would be no need to build a new infrastructure for last-mile access. So wherever electricity is available, there could be broadband.

In this perspective, broadband over power lines (BPL) technology seems to emerge as a new business prospective for established telecom operators, utility companies and system manufacturers to provide high-speed broadband services that can cover each and every home or office due to almost omnipresence of power lines. With no need for new wiring or major infrastructure deployment, BPL creates an alternative broadband solution that could lead to lower prices for broadband consumers. Thus it is hoped that BPL has the potential to become an effective means for last-mile delivery of broadband services and may offer a competitive alternative to other high-speed Internet access technologies.

BPL building blocks

BPL is a power-line communication technology that allows high-speed

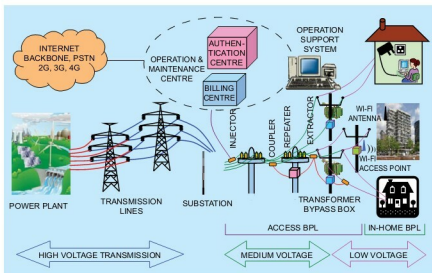


Fig. 1: BPL building blocks

digital data transmission over utility power lines. By utilising the combination of technological principles of radio, power engineering, electromagnetic compatibility, networking and modem technology, it offers instantaneous access to high-speed Internet from in-house electrical outlet. In order to access broadband services, subscribers need to install a modem that plugs into an ordinary wall outlet.

BPL systems use the existing electrical power lines as a transmission medium to provide high-speed telecommunications capabilities by coupling radio frequency (RF) energy onto the power line. In order to carry high-speed data, BPL uses radio spectrum ranging from high-frequency (3-30 MHz) to lower portion of very high-frequency (30-300 MHz) allocated to other over-the-air communication services. Because the electric current

(50 Hz) and RF (typically 2-80 MHz for BPL) have different frequencies, the two don't interfere with each other.

There are two predominant types of BPL communication configurations: Access BPL systems that couple RF energy onto medium-voltage (MV) and low-voltage (LV) power lines; and in-home BPL networks, which use existing electrical outlets available within a home or office for the provision of a local-area network (LAN). A typical BPL network set-up is shown in Fig. 1.

The existing three-tiered power grid hierarchy, which comprises HV, MV and LV transmission lines, is exploited to carry the high-speed digital data over a broad range of frequencies without causing any significant interference to the rightful incumbent users of those frequency bands. The electricity is generated from power plants (thermal, hydro, nuclear, wind turbines or solar),